

An analytical approach to credit risk assessment using machine learning models

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ABSTRACT

This study presents a novel Early Warning System for monitoring the credit risk of commercial customers at a large international bank headquartered in the Netherlands. Traditional early warning methods often rely on backward-looking indicators such as probability of default or loss given default, which can limit predictive performance. To address this, we investigate the effectiveness of a Watchlist-based trigger for forecasting financial distress and adverse customer migration. We assess its precision, timeliness, and sensitivity across different client status transitions. Using a rich dataset combining internal banking records and external financial information, we implement and compare several machine learning algorithms, including Linear Discriminant Analysis, Logistic Regression, Decision Trees, Support Vector Machines, Random Forest, Gradient Boosting, Extreme Gradient Boosting, and Artificial Neural Networks. To enhance model transparency and support adoption, we employ SHapley Additive exPlanations to interpret key predictors of risk. Among all models, Random Forest achieves the highest performance, demonstrating strong F1 scores, superior trigger precision, and high sensitivity to migration. It successfully anticipates 12.7% of negative client transitions and helps prevent 67.6% of cases that would otherwise result in financial losses for the bank. This research contributes a data-driven, explainable solution for proactive credit risk management and offers actionable insights to support strategic decision-making in commercial banking.

1. Introduction

The proactive detection of credit risk is important for identifying consumers facing financial challenges early and mitigating potential losses effectively. The implementation of Early Warning Indicators (EWI) and triggers enables risk and account managers to diligently monitor their credit portfolios, thereby playing a crucial role in minimizing exposure for both retail and commercial clients [1]. Moreover, Early Warning Systems (EWS) are instrumental in strengthening customer relationships by offering timely guidance to those in distress, potentially averting bankruptcy. The necessity of EWS extends to regulatory compliance, compelling banks to substantiate any deviations [2].

In the contemporary banking landscape, the integration of big data into the identification of credit risk emerges as a critical factor for maintaining competitiveness. Traditional early warning models, which predominantly analyzed accounting variables, are now being enhanced with diverse data sources, thanks to advancements in internet capabilities and computational power. Those changes allow for the incorporation of real-time financial data and online news articles, thereby enriching the capabilities of early warning systems.

The application of Machine Learning (ML) techniques in managing vast data streams has been extensively documented in the literature. For instance, Liang et al. [3] demonstrate how distributed data infrastructure and optimized random forest algorithms can be employed to assess credit risk in Internet of Things (IoT) finance, reducing misclassification errors and improving banks' profitability. Similarly, Fang and Zhang [4] highlight how big data technologies support algorithmic trading and risk management in quantitative finance by enabling real-time processing, visualization, and integration of high-volume, high-velocity, and high-variety data streams. In addition, Machado and Karray [5] show that hybrid ML approaches – combining supervised and unsupervised learning – can significantly enhance credit score prediction for commercial customers. Their findings suggest that incorporating past credit scores, often overlooked in prior research, further improves predictive accuracy. These results underscore the potential of integrated ML frameworks over traditional statistical models in credit scoring tasks. These studies have been demonstrating how ML applications have been changing the way banks detect and manage credit risk. Traditionally, banks relied on ad hoc triggers based on single financial metrics, which often led to imprecise models characterized by high

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false positive rates. By leveraging ML, banks can automate decision-making processes and integrate a multitude of variables, significantly enhancing the accuracy of early warning models. This paper focuses on how a commercial bank can employ ML models to effectively identify commercial clients at risk of distress using EWS data as input in the prescriptive frameworks.

In the banking sector, serving both retail and commercial clients, the continuous monitoring of loan portfolios is crucial due to the changing risk profiles of clients. The development of advanced EWS represents a significant leap forward. These systems utilize a wide range of data sources, both internal and external, to preemptively identify potential financial risks. Their ability to analyze large volumes of data reflects a comprehensive approach to credit risk management, using sophisticated models to detect early signs of critical corporate events, such as fraud or bankruptcy.

This research aims to build upon the current capabilities of EWS by evaluating the effectiveness of specific early warning triggers in managing credit risk. It proposes the introduction of a Watchlist (WL) trigger, designed to more accurately classify clients who are at risk of financial distress. By implementing an ML approach, this study seeks to enhance the EWS's accuracy. The expected result is a reduction in the dependence on manual processes for the classification of at-risk clients, leading to a more efficient, consistent, and precise assessment of client risk profiles.

The core research question explored is: *"How can a WL trigger be optimally designed to accurately identify and categorize commercial clients based on their forecasted credit risk, thereby enhancing the effectiveness of credit risk management practices?"* This study analyzes the relationship between early warning triggers and watchlist status, develop a financial distress prediction model, and implement a new WL trigger, evaluating its integration into current industry practices.

In addressing this research question, the study makes several important contributions to the field of financial distress prediction for commercial clients:

- **Innovative Analysis of Early Warning Triggers (EWTs):** The study evaluates the effectiveness of existing EWTs using migration sensitivity, trigger precision, and time lag as key metrics.
- **Development of a Comprehensive Predictive Model:** A machine learning model is introduced that integrates historical trigger events with internal customer data, with the Random Forest algorithm demonstrating particularly strong predictive performance.
- **Introduction of a Novel Watchlist (WL) Trigger:** A new WL trigger is proposed, which outperforms existing triggers in terms of early detection capabilities.
- **Extension of the Financial Distress Concept:** The research broadens the definition of financial distress to include watchlist classifications, allowing for earlier and more nuanced risk identification.
- **Enhanced Model Interpretability through SHAP Values:** The study improves transparency and explainability of the predictive model by applying SHAP values to identify key drivers of risk predictions.
- **Practical Case Study Implementation:** Conducted in collaboration with a major Dutch commercial bank, the research bridges theoretical insights with real-world application, laying the groundwork for future improvements in early warning systems and financial risk management.

The rest of the paper is organized as follows: Section 2 discusses the literature relevant to this investigation. Sections 3 and 4 outline the methodologies and the experimental settings. Section 5 contains the results and discussion, while Section 6 provides the conclusion.

2. Literature review

This section reviews the existing literature on the application of machine learning in credit risk assessment, with a particular focus on early warning systems (EWS) for financial distress. We examine prior work on early warning monitoring frameworks, highlighting key methodological trends and identifying gaps that motivate our study.

2.1. ML in credit risk

Credit risk management is an essential activity in the banking sector, involving the assessment of potential losses when borrowers fail to fulfill their financial commitments [6]. This assessment is key for a bank's profitability, regulatory compliance, and competitive edge [7]. ML emerges as a powerful tool in this context, capable of handling complex, non-linear relationships, integrating vast datasets for decision-making, and automating lending processes. For instance, Aziz and Dowling [8] provide a comprehensive overview of how ML and AI techniques are transforming risk management across domains such as credit, market, and operational risk, while also highlighting current limitations such as data governance and skill gaps. Lahmiri et al. [9] apply various ML models to house price prediction and find that boosting ensemble regression trees significantly outperform both traditional regression and neural networks, showcasing the power of ensemble methods in financial forecasting tasks. Further, Razaque et al. [10] propose a novel predictive-based reinforcement learning (PRL) framework that combines reinforcement learning with explainable AI (LIME and SHAP) to dramatically improve credit scoring accuracy for manufacturers and importers, achieving over 99.5% accuracy and demonstrating superior performance over existing approaches. van Braak et al. [11] address fairness in credit risk prediction for marginalized consumers by integrating ensemble learning with bias mitigation tools like the disparate impact remover, showing how careful handling of imbalanced data and fairness trade-offs can maintain strong predictive performance. Lastly, Das et al. [12] offer a comprehensive review of combining sentiment analysis with graph neural networks (GNNs) for stock prediction, emphasizing how novel data structures and unstructured data (e.g., news and social media) can be harnessed to boost forecasting accuracy and provide actionable insights in financial markets.

These application of ML in managing credit risk spans three main areas: fraud detection, credit scoring, and financial distress prediction [13]. Fraud detection models utilize anomaly detection techniques and transaction history to identify patterns indicative of fraud [14]. Credit scoring models evaluate the creditworthiness of clients, aiding banks in decision-making about loan approvals or rejections based on supervised ML methods, which have been extensively explored in research [7,15,16]. The third area, financial distress prediction, employs ML to foresee when a client may face financial difficulties, allowing banks to intervene proactively. While credit scoring and financial distress prediction both involve binary classification, they differ significantly in their implications for false positives. In credit scoring, false positives can lead to declined loans and mispricing, whereas in EWS, they primarily increase staff workload without necessitating immediate action [17]. This research focuses on the latter, particularly on the utilization of ML models for predicting financial distress, which will be elaborated upon in the following sections.

2.2. Early warning monitoring of financial distress

EWSs serve as critical tools in the financial industry for identifying potential risks at their nascent stages, enabling timely preventive or mitigative measures to avert adverse outcomes [18–21]. These systems, equipped with triggers and indicators, alert stakeholders to early risk signs, thus facilitating prompt action [1]. The versatility of EWS is evident in its wide-ranging applications, from forecasting financial crises by central banks to monitoring the credit risk of sovereign debts

and identifying banks at risk of failure [22–24]. Specifically, in the context of credit risk, EWS plays a substantial role in safeguarding lenders and other stakeholders from potential defaults [25].

This research examines the application of financial distress prediction models within the framework of EWS, with a particular focus on commercial clients. A comprehensive literature review has been conducted to compare various aspects of these models, including their predictive approaches, methodologies and algorithms used to forecast the likelihood that a company will face financial difficulties in the future, independent variables used, and the definitions of financial distress employed. Additionally, the application of these models in EWS and the client types covered are scrutinized.

One significant challenge identified in the literature is the lack of a uniform definition of financial distress, leading to diverse dependent variables for model training. While some models rely on bankruptcy data to identify distressed clients [26], others advocate for a broader interpretation of financial distress. This broader view encompasses both the early and advanced stages of distress, aiming to enable early interventions that could mitigate higher costs associated with late-stage financial distress [27,28].

The criteria for defining financial distress have been expanded in the literature, with various studies developing prediction models based on data from the China Securities Regulatory Commission (CSRC), which identifies companies in financial jeopardy after consecutive years of losses [29]. Other approaches to labeling companies as distressed include evaluating cash flow thresholds [30] and assessing whether accumulated losses exceed the net worth of a company [31]. Multiple criteria, including book value and dividend declarations, have also been utilized to define financial distress [27].

The predictive models for financial distress are predominantly based on supervised machine learning techniques, employing a wide array of algorithms from Decision Trees (DT) and Logistic Regression (LR) to advanced ensemble methods like Random Forest (RF) and XGBoost [18,28,32]. Additionally, traditional financial analysis techniques, such as the Altman Z-score, and traditional approaches like the KMV model, which assesses credit risk through option pricing, have been implemented [27,29]. Some models also incorporate expert judgment through Multiple-Criteria Decision Analysis (MCDA) or Fuzzy Logic [17,19,33].

To enhance the accuracy of financial distress predictions, recent studies have broadened the scope of data inputs beyond traditional financial statements. These include market data, non-financial indicators like company age and governance characteristics, and textual data through Natural Language Processing (NLP) techniques [28,34,35]. Such innovations aim to provide a more holistic view of a company's financial health and potential distress signals.

2.3. Early warning systems

While financial distress prediction models have been extensively studied, their integration into EWS remains underexplored. The existing literature discusses primarily the development of financial distress prediction models without going into their applications within EWS frameworks. However, a few studies have broadened the scope of this discussion. For instance, [32] analyzes into the data architecture necessary for harnessing big data in EWS, while [18] outlines a data flow framework designed to help clients improve their risk profiles. Notably, research conducted by [17] stands out as a unique instance where an EWS incorporating financial distress prediction based on Fuzzy Logic was implemented at an individual financial institution. This approach included a univariate analysis of early warning triggers, integrating these triggers into the model. Furthermore, [36] develop a data-driven EWS tailored for retail customers in the financial industry, addressing a gap in existing applications which have traditionally focused on commercial clients. They apply multiple ML models – Logistic Regression, Gradient Boosting, and Random Forest – to classify customer credit

risk and propose a meta-model that integrates the outputs of these individual classifiers. Their findings show that the meta-model outperforms standalone models in predictive accuracy. They also explore how varying warning thresholds affect the timing and sensitivity of alerts, and demonstrate that using more than the top ten features can enhance decision-making. Similarly to Beltman et al. [36] this study applies EWS for credit risk assessment, however, in the for commercial (wholesale) settings.

This gap in the literature highlights a lack of detailed exploration into the application of ML techniques for financial distress prediction at the individual financial institution level. Addressing this void, the present study undertakes a case study at an international commercial bank headquartered in the Netherlands. Building upon the groundwork laid by Kaludher and Klepac [17], this research further investigates the relationship between early warning triggers and the watchlist classification status of clients. It introduces a novel financial distress prediction model that employs a distinct dependent variable for forecasting financial distress and considers various data sources for independent variables. Unlike the predominant focus on publicly listed companies in existing ML models, this study pioneers the implementation of these models at the individual institutional level, leveraging internal data. This approach not only seeks to refine the predictive accuracy of financial distress models but also to enhance the operational efficacy of EWS within the banking sector.

3. Methodology

This section outlines the methodological framework used in our study, including the supervised machine learning models employed for financial distress prediction. We also detail the evaluation metrics used to assess model performance and describe the application of explainable AI (XAI) tools to enhance model transparency and interpretability.

3.1. Supervised learning

Supervised learning, the primary approach identified in the literature for predicting financial distress, involves training models on datasets where the target outcomes are already known [37]. In this context, classification models are particularly suitable for the early detection of financial distress. The models most frequently adopted in this area include Linear Discriminant Analysis (LDA), Logistic Regression (LR), Support Vector Machines (SVM), Decision Trees (DT), ensemble learning methods, and Artificial Neural Networks (ANNs).

LDA, for instance, seeks a linear combination of predictors that best separates different classes, aiming to maximize the variance between categories while minimizing the variance within them [38]. Its well-known application in financial distress prediction is Altman's Z-score model, which incorporates five financial ratios to predict bankruptcy [39].

Logistic Regression is another widely adopted model for binary classification tasks. It estimates the probability of class membership using the logistic function and optimizes weights through Maximum Likelihood Estimation. Its simplicity and efficiency make it highly popular in financial distress prediction studies, with multiple application cases presented in the literature [34,40,41].

SVMs, introduced by Boser et al. [42], identify a hyperplane that maximizes the margin between classes and are capable of handling non-linear separability through kernel functions. Their ability to generalize and manage high-dimensional data makes them useful for classification tasks related to financial distress.

DT models operate by recursively partitioning the feature space, guided by purity measures such as entropy or Gini impurity [43]. Their interpretability and ease of explanation have led to their widespread use in financial distress studies [30,41,44]. Measures like pruning or limiting tree depth are used to prevent overfitting [45].

Ensemble learning methods, such as bagging and boosting, aggregate multiple models to enhance predictive performance and robustness [46]. Bagging, implemented through methods like Random Forests (RF), reduces model variance by training on bootstrapped samples [47, 48]. Boosting, on the other hand, sequentially emphasizes misclassified instances and has inspired powerful techniques such as AdaBoost, Gradient Boosting (GB), CatBoost, LightGBM, and XGBoost [49].

Finally, ANNs simulate neural structures through interconnected layers of nodes, or "neurons", which process inputs through weighted connections and activation functions [50]. Model training is achieved through backpropagation, optimizing weights to minimize error via algorithms like Gradient Descent. ANNs are known for their ability to model complex, non-linear relationships.

3.2. Evaluation metrics

To evaluate the performance of models predicting financial distress, a set of well-established metrics are derived from the confusion matrix. This matrix categorizes outcomes as true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), thereby offering a structured basis for performance assessment.

Accuracy is a commonly reported metric, reflecting the ratio of correctly predicted observations to the total number of predictions [27,31, 51,52]. However, in cases of class imbalance, such as when distressed firms represent a small minority, accuracy alone can be misleading. For example, a model predicting no financial distress in a dataset with only 0.8% distressed cases could still achieve 99.2% accuracy without identifying any actual distress.

The Receiver Operating Characteristic (ROC) curve and its Area Under the Curve (AUC) offer more nuanced insights. The ROC plots the True Positive Rate (TPR, or sensitivity) against the False Positive Rate (FPR) at various thresholds [53], and the AUC summarizes this discrimination ability across all thresholds [34,54,55]. While AUC is robust to class imbalance, it may still give an overly optimistic view in highly skewed datasets.

To complement these, the F1 score and the Matthews Correlation Coefficient (MCC) are increasingly adopted to address class imbalance challenges in financial prediction tasks. For instance, Huang et al. [34] demonstrate that optimizing MCC directly leads to superior fraud detection performance compared to traditional metrics, particularly in highly skewed datasets. Sreedharan et al. [56] introduce a multi-objective ensemble model for credit classification, explicitly using F1 and MCC as optimization targets to balance sensitivity and specificity. Similarly, Ribeiro et al. [57] evaluate credit scoring models under imbalanced settings and find that MCC provides a more reliable assessment than accuracy, especially when misclassification costs are asymmetric. Hájek and Olej [58] apply various performance metrics in bankruptcy prediction, concluding that MCC and F1 yield more stable results under severe class imbalance. Finally, Pavlicko et al. [59] explore ensemble-based classifiers for financial distress prediction and highlight the advantages of using MCC to guide model selection and evaluation in real-world, imbalanced datasets. The F1 score, which is the harmonic mean of precision and recall, balances the trade-off between identifying all positive instances and minimizing false positives [60]. MCC further evaluates the quality of binary classifications by considering all four elements of the confusion matrix, making it particularly robust in imbalanced scenarios.

Precision and sensitivity are also critical when the focus lies on correctly identifying the positive class—i.e., financially distressed firms. Precision measures the proportion of correctly identified positives among all predicted positives, while sensitivity measures the ability to detect actual positives [34,61,62]. Together, these metrics ensure that models are not only accurate in general but also reliable and informative when it comes to identifying distressed firms—an essential goal in risk-sensitive domains like finance.

3.3. Model explainability

Explainable AI (XAI) encompasses techniques and methodologies aimed at rendering the decision-making processes of ML models transparent and comprehensible to humans [63]. The importance of XAI in financial distress prediction modeling cannot be overstated for several reasons. Firstly, it enables risk managers to elucidate to clients the rationale behind their inclusion on a watchlist, a factor with significant implications for their perceived creditworthiness and their ongoing relationship with the bank. Secondly, the explainability of models facilitates the identification of factors leading to watchlist classification, thus guiding appropriate interventions. Lastly, XAI contributes to the validation of model predictions, bolstering confidence in the models' decision-making capabilities.

SHAP (SHapley Additive exPlanations) values, drawing on concepts from cooperative game theory as introduced by [64], offer a robust method for quantifying the influence of individual features on the output of an ML model. In this framework, features are likened to players in a game, with their contributions to the model's output being analogous to a player's contribution to the game's outcome. The calculation of SHAP values involves assessing the impact of including versus excluding a feature across all possible feature subsets, with the output differences being weighted according to the number of feature combinations that incorporate the given feature. Through aggregating these weighted contributions for each feature across all subsets, SHAP values effectively quantify the change in the model's prediction attributable to the inclusion of each feature.

4. Experimental setup

This section describes the experimental setup used to implement and validate our predictive framework. We detail the dataset and preprocessing steps, the development of machine learning models, and the procedures followed for performance evaluation.

4.1. Background & data

This study constructs a watchlist classification model utilizing a financial distress prediction approach that leverages historical data on early warning triggers. The design of this model is guided by specific requirements identified through problem analysis and exploratory data analysis, summarized as follows:

- **Negative Migrations:** The focus is on predicting negative migrations, defining financial distress as a client's transition to watchlist or default status. While positive migrations are not considered, their inclusion represents a potential avenue for future research.
- **Sequentiality:** The model aims to forecast future migrations using historical data, necessitating features that reflect the temporal sequence of events.
- **Earliness:** The goal extends beyond pinpointing the exact timing of a migration; early detection of financial distress, allowing for preemptive action, is deemed more valuable.
- **Error Costs:** Balancing the costs of false positives against false negatives is essential, with the latter posing a greater risk due to the missed opportunity to prevent defaults.

Fig. 1 illustrates the early warning trigger history for a hypothetical client, highlighting the detection of four triggers over twelve months leading to a watchlist classification. This historical insight underpins the model's predictive endeavor, which seeks to forecast watchlist or default status within a one-month future window. This predictive framework introduces a target window concept, not strictly focusing on the exact migration moment but rewarding early detection within a five-month pre-migration period.

The development and evaluation of the financial distress prediction model encompass data preprocessing, model construction, and

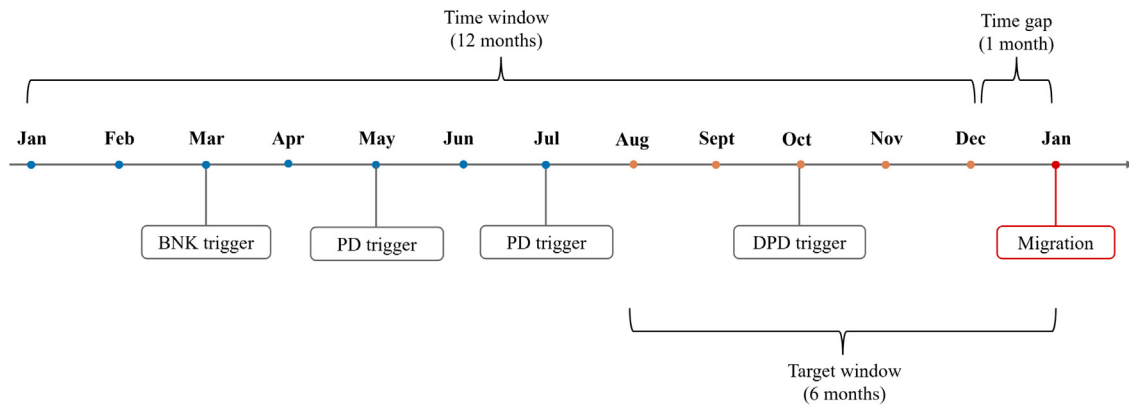


Fig. 1. Example of early warning trigger history of a client in financial distress.

performance assessment. Fig. 2 outlines the procedural flowchart for these phases. The process begins with foundational data preparation steps, including merging various data sources, imputing missing values, labeling the target variable, engineering relevant features, filtering the dataset, and splitting it into training and testing sets. This is followed by model development tasks such as data scaling, addressing class imbalance, selecting the most relevant features, choosing appropriate algorithms, and tuning their hyperparameters. Finally, the workflow concludes with generating predictions, validating model performance, extracting evaluation metrics, and interpreting results using explainable AI techniques.

The computational complexity of this study is driven by the extensive data preprocessing pipeline, high-dimensional feature space, and the iterative nature of model development. The construction of lagged features across a six-month historical window, combined with one-hot encoding and feature engineering, expanded the dataset to 198 features and over half a million monthly records—requiring substantial memory and processing capacity. Moreover, the use of Random under-sampling to address severe class imbalance and mutual information-based feature selection added additional layers of processing. Model training further contributed to computational demands, as multiple supervised learning algorithms were evaluated with hyperparameter tuning performed via random search, followed by expanding-window forward cross-validation tailored for temporal data. These processes involved repeated model training and evaluation across sequential time splits, making them computationally intensive. To manage this complexity efficiently, all experiments were executed using Python 3.9 in Jupyter Notebooks with TensorFlow on a high-performance machine running Ubuntu 20.04, equipped with a 16-core (32-thread) Intel Xeon Silver 4216 CPU and 96 GB of RAM. This infrastructure enabled scalable experimentation while maintaining reproducibility and robustness.

4.2. Data preprocessing

Although in this section, we provide a detailed description of the data preprocessing steps, it is important to note that the dataset used in this study consists of confidential internal data from a major Dutch commercial bank. Due to privacy and security agreements, the specific features and their values cannot be disclosed. Nevertheless, the preprocessing methodology – including data integration, imputation, feature engineering, labeling, and filtering – is thoroughly documented to support transparency and replicability. Researchers or practitioners working with similar datasets or facing comparable challenges in financial distress prediction can follow the outlined steps and adapt the proposed framework to their context, ensuring the broader applicability of our approach despite the proprietary nature of the original data.

Furthermore, this research's financial distress prediction model undergoes various preprocessing stages, including data integration, imputation, feature engineering, classification labeling, and data filtering, as outlined below:

- **Data Integration:** Utilizing three datasets – internal client information and both internal and external trigger records – this phase involved combining client financial details with trigger event data. Due to missing monthly records, client status information was manually sourced. A binary scheme was applied to categorize the presence or absence of triggers, subsequently merged with client data using unique identifiers.
- **Imputation:** Missing values were addressed next, with unrecorded trigger events assumed absent and imputed as zero. Internal data gaps were filled with the last known value or, if unavailable, the overall mean or mode for numerical or categorical data, respectively.
- **Feature Engineering:** Involved transforming categorical variables via one-hot encoding and generating lagged features to capture historical data trends. A six-month window and a one-month lag were selected based on a balance between historical detail and sample size. This process expanded the feature count from 30 to 198, solely based on the use of one hot encoding and lag features (see Tables 1 and 2).
- **Data Labeling:** The binary outcome variable was created based on client status transitions, defining financial distress as any negative migration. A target window approach was employed (see Tables 2), labeling the period prior to a negative migration as indicative of distress.
- **Data Filtering:** Non-relevant data points were excluded, focusing on data between January 2021 and September 2022 due to the availability of comprehensive trigger data. Triggers lacking a year of historical data were omitted, and despite initial considerations, data related to Russian clients was retained.

This preprocessing strategy leads to a dataset encompassing 548,210 monthly records across 38,316 clients.

4.3. Model development

The model development constructs a model for predicting financial distress, addressing challenges such as class imbalance, data normalization, feature relevance, algorithm selection, and hyperparameter optimization.

- **Class Imbalance Management:** Given the predominance of unchanged client statuses, resulting in a mere 0.8% of data representing negative migrations, Random Under Sampling (RUS) was employed to equilibrate class distribution by randomly downsizing the majority class. Despite potential information loss, RUS offers computational efficiency by reducing dataset size, which is crucial for model training. The optimal minority-to-majority class ratio was determined through hyperparameter tuning.

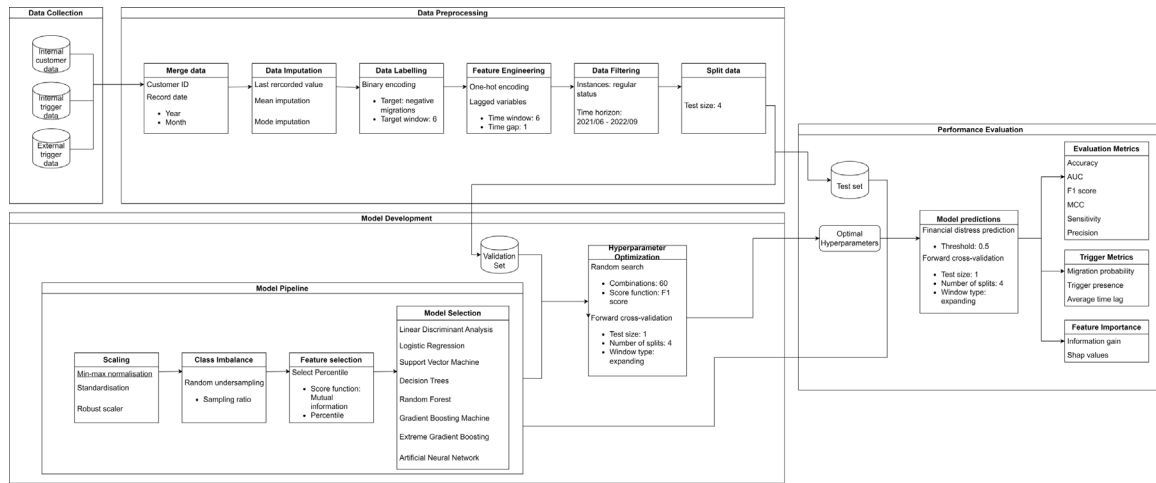


Fig. 2. Flow chart of the financial distress prediction model.

Table 1
Example of lagged variables.

Trigger/lag	Lag 12	Lag 11	Lag 10	Lag 9	Lag 8	Lag 7	Lag 6	Lag 5	Lag 4	Lag 3	Lag 2	Lag 1
PD	0	0	0	0	1	0	1	0	0	0	0	0
BNK	0	0	1	0	0	0	0	0	0	0	0	0
DPD	0	0	0	0	0	0	0	0	0	1	0	0
LGD	0	0	0	0	0	0	0	0	0	0	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...

Table 2
Example label encoding (R: regular status, W: watchlist status, D: default status).

	2021/12	2022/01	2022/02	2022/03	2022/04	2022/05	2022/06	2022/07	2022/08	2022/09
Client status	R	R	R	R	R	W	W	W	D	D
Migration (from, to)	n.a.	(R, R)	(R, R)	(R, R)	(R, R)	(R, W)	(W, W)	(W, W)	(W, D)	(D, D)
Label encoding	n.a.	0	0	1	1	1	0	0	0	0

Table 3
Expanding-window forward cross-validation overview.

2021/06	...	2021/10	2021/11	2021/12	2022/01	2022/02	2022/03	2022/04	2022/05	2022/06
Training			Validation							
Training				Validation						
Training					Validation					
Training						Validation				
Training							Validation			
Training								Test		
Training									Test	
Training										Test
Training										

- **Data Normalization:** Min-Max Normalization was selected from among various scaling techniques due to its performance and the interpretability of normalized trigger-based features. This method adjusts numerical data into a 0–1 range, enhancing model comparability and performance.
- **Feature Selection Using Mutual Information:** To tackle the explosion in feature quantity from lagged feature creation, mutual information was utilized for feature selection, identifying the most predictive variables by measuring the shared information between features and the target variable. This method is advantageous for its applicability to both continuous and discrete data, its ability to capture non-linear dependencies, and its computational efficiency.
- **Model Selection and Supervised Learning:** A subset of supervised learning algorithms identified in literature – LDA, LG, DT, SVM, RF, GBM, XGB, and ANN – was tested alongside a baseline dummy model for benchmarking. The ANN models explored included two and three-layer architectures, employing ReLU activation and dropout techniques to mitigate overfitting,

with architecture specifics refined through hyperparameter optimization.

- **Hyperparameter Optimization via Random Search:** Random search, preferred for its computational efficiency over grid search, was used to identify the best hyperparameter combinations for each model, focusing on achieving the highest F1 score to accommodate class imbalance. [Table A.1](#) in the Appendix show the hyperparameters we use in each model.
- **Model Validation with Expanding-Window Forward Cross-Validation:** To ensure robustness and mitigate overfitting, models were validated using an expanding-window forward cross-validation approach suitable for sequential data ([Table 3](#)). This method preserves the chronological order of data, distributing negative migrations evenly across folds to maintain class balance within validation and testing phases.

4.4. Performance evaluation

The evaluation of the model's efficacy is twofold: as an ML predictive tool and as an EWS trigger. For the ML aspect, standard metrics are utilized to analyze performance, including:

- **Accuracy:** Overall correctness of the model.
- **Area Under the Curve (AUC):** Reflects the model's ability to distinguish between classes.
- **F1 Score:** Balances precision and recall, particularly useful in imbalanced datasets.
- **Matthews Correlation Coefficient (MCC):** Offers a comprehensive measure of the model's quality.
- **Sensitivity:** Measures the model's capability to identify positive cases.
- **Precision:** Assesses the accuracy of the positive predictions made by the model.

Additionally, the model is compared against existing EWS triggers to ascertain if it enhances the identification of financially distressed clients at an earlier stage, employing metrics such as migration sensitivity, trigger precision, and time lag for this comparative analysis.

We also exploit the use of the model's explainability with SHAP Values. Emphasizing SHAP (SHapley Additive exPlanations) values, this research aims to demystify the watchlist classification process. SHAP values offer a detailed view of feature importance by quantifying each feature's contribution to the model's output. This model-agnostic technique enables its application across various ML models, providing insights into the decision-making process. The adoption of SHAP values for explainability is well-documented within the credit risk field, underscoring its relevance and utility in enhancing transparency and trust in ML predictions.

5. Results & discussions

The results and discussions of this study are structured around a series of experiments designed to address the main research questions, examining the effectiveness of various supervised learning models in predicting negative migrations. These experiments explore different aspects of model performance and configuration:

- E1: Evaluates which supervised learning model most accurately predicts negative migrations.
- E2: Compares the model's ability to predict negative migrations against its performance on predicting default and watchlist migrations individually.
- E3: Investigates the impact of different time window sizes on model performance.
- E4: Assesses how variations in the target window size affect model accuracy.
- E5: Examines the relationship between time gap size and prediction accuracy.

Initially, the study assesses the performance of prevalent models in financial distress prediction, including LDA, LG, DT, SVM, RF, GBM, XGB, and ANN, under a standardized configuration to ensure comparability. This baseline configuration—comprising a six-month time window, one-month time gap, and six-month target window—balances the need for sufficient training data with computational efficiency.

The analysis further differentiates between watchlist and default migrations to ascertain the model's predictive capacity across distinct migration types. The exploration into the time window's influence on performance considers the potential for enhanced accuracy through increased historical data, mindful of the trade-off between the window size and the available training dataset size. Similarly, the study probes the relationship between target window size and model timeliness, hypothesizing improved performance with greater target window sizes but also anticipating potential increases in the average prediction time lag.

Lastly, the effect of time gap size on model accuracy is scrutinized, with the expectation that larger gaps might challenge the model's predictive capability, underscoring the necessity to balance prediction earliness against overall performance.

Table 4

Evaluation metrics for each classifier.

Model	F1 score	Accuracy	AUC	Precision	Recall	Correlation
DT	0.073	0.911	0.688	0.044	0.231	0.069
LDA	0.069	0.866	0.646	0.040	0.295	0.068
LG	0.098	0.936	0.646	0.063	0.228	0.093
SVM	0.077	0.927	0.648	0.049	0.203	0.070
GBM	0.096	0.935	0.737	0.063	0.230	0.092
RF	0.197	0.930	0.893	0.119	0.582	0.240
XGB	0.161	0.914	0.879	0.095	0.561	0.203
ANN2	0.050	0.963	0.673	0.042	0.064	0.033
ANN3	0.055	0.961	0.667	0.043	0.082	0.040
DUMMY	0.029	0.499	0.500	0.015	0.488	-0.003

5.1. Predictive performance

Optimal hyperparameters are determined through model adjustment for every experiment design. Fig. 3 shows the maximum F1 score achieved by each machine learning model using their best hyperparameters. The F1 scores do not accurately reflect the models' performance as it must be evaluated using unseen data. Experiments are limited to the best-performing model due to time constraints. The evaluation of performance is done using the validation set to prevent bias. RF exhibits superior performance in this scenario. Hence, the Random Forest model is utilized to assess the effectiveness of various migration strategies, time frames, and target periods.

Furthermore, mutual knowledge was utilized to choose the most essential aspects for the model. The mutual information suggests that variables derived from internal customer data exhibit the most significant correlation with bad migrations, but early warning triggers show a comparatively less association. The models will primarily include characteristics derived from internal data following feature selection. It is more advantageous to incorporate the underlying data for triggers rather than the triggers themselves. One drawback of mutual information is its failure to account for the interaction effects of features on predictive performance. Hence, the SHAP values will be analyzed in the following section to assess the collective influence of characteristics on model performance.

Ten classifiers were assessed using six evaluation criteria, as displayed in Table 4. The RF model outperformed the other models with an F1 score of 0.197, an AUC of 0.893, and an accuracy and sensitivity of 0.119 and 0.582, respectively. The results show that the RF model can differentiate between positive and negative instances, accurately identifying a significant number of true positives while reducing false positives. Additional ensemble models that showed satisfactory performance are GBM and XGB, with F1 scores of 0.096 and 0.161, respectively. Nevertheless, these models exhibited reduced precision and sensitivity compared to the RF model, leading to inferior overall performance.

The Decision Tree, Linear Discriminant Analysis, Support Vector Machine, and Logistic Regression models exhibited inferior F1 scores, accuracies, and AUC values in comparison to the top-performing models. Artificial Neural Networks (ANNs) with two and three hidden layers exhibited low F1 scores of 0.050 and 0.055, respectively, suggesting inadequate performance. This is likely a result of overfitting, as they demonstrated good performance on the validation set. Each classifier outperformed the dummy model, demonstrating superior performance compared to random.

In general, the results obtained in this study align with findings reported in the literature for similar contexts. The accuracy achieved in [36], which also proposes an Early Warning System (EWS) for credit risk assessment, ranges between 65% and 90%, consistent with the accuracies observed in our study. Similarly, other studies on credit risk assessment – though not EWS-based – using comparable machine learning models (e.g., tree-based models and neural networks) report accuracies between 70% and 90% [5,11], further supporting the validity of our results.



Fig. 3. Model validation F1 score.

Table 5

RF model performance for different target variables.

Prediction approach	F1-score	Accuracy	AUC	Precision	Sensitivity	Correlation
Negative migration	0.197	0.930	0.893	0.119	0.582	0.240
Default migration	0.151	0.949	0.955	0.083	0.827	0.251
Watchlist migration	0.120	0.911	0.897	0.067	0.646	0.187

Table 6

RF model performance for different time window sizes.

Time window	F1 score	Accuracy	AUC	Precision	Sensitivity	Correlation
2	0.184	0.927	0.882	0.112	0.555	0.224
4	0.179	0.929	0.882	0.109	0.523	0.214
6	0.197	0.930	0.893	0.119	0.582	0.240
8	0.220	0.934	0.906	0.135	0.628	0.268
10	0.234	0.935	0.919	0.142	0.671	0.288

5.2. Early warning (negative migrations)

The strategy that concentrated on forecasting negative migrations achieved the greatest F1 score of 0.197, surpassing the default migration and watchlist migration models, which scored 0.151 and 0.120, respectively (Refer to Table 5).

The default migration as the goal variable showed improved sensitivity compared to the negative migration strategy, indicating fewer false negatives in the model. Additionally, the default migrations are more predictable than watchlist migrations since they achieve higher scores across all evaluation metrics. This could be due to the fact that watchlist classifications rely on expert judgment, which may result in inconsistencies in the data.

5.3. Different time windows

The Table 6 presents the evaluation metrics for various time windows. The model's performance improves as the time windows expand. The decrease in performance is presumably due to the reduced amount of data available for training the model. We believe that the model could achieve improved performance by utilizing additional historical data. It would be intriguing to explore the ideal time frame size with increased data availability.

5.4. Target window analysis

A one-month goal window indicates that the model aims to estimate the specific month when a negative migration will take place. The data in Table 7 indicates that the model's performance in forecasting financial distress is subpar. Therefore, the goal window was implemented to enhance the model's flexibility by enabling earlier predictions. This aligns with our objective as we strive to identify financial distress at the earliest opportunity. Extending the target window greatly improves the model's performance, as indicated by the results. Due to data

Table 7

RF model performance for different target window sizes.

Target window	F1 score	Accuracy	AUC	Precision	Sensitivity	Correlation
1	0.037	0.959	0.694	0.020	0.197	0.051
2	0.074	0.934	0.769	0.041	0.342	0.100
4	0.156	0.927	0.857	0.092	0.514	0.194
6	0.197	0.930	0.893	0.119	0.582	0.240

Table 8

RF model performance for different time gap sizes.

Time gap	F1-score	Accuracy	AUC	Precision	Sensitivity	Correlation
1	0.197	0.930	0.893	0.119	0.582	0.240
2	0.201	0.931	0.894	0.122	0.586	0.244
3	0.212	0.937	0.900	0.131	0.572	0.251
4	0.214	0.930	0.905	0.132	0.628	0.262

constraints, only a goal window of up to six months was examined. However, with additional migration data, it would be intriguing to study the most effective target window.

Additionally, these measurements may underestimate the models' functional performance as a catalyst. These metrics indicate the model's ability to accurately forecast all negative migrations during the specified time period. In our situation, the predictions are adequate if they can identify a negative migration as soon as possible.

5.5. Time gap analysis

We predicted that the time interval would adversely affect the model's performance, however this does not appear to be true based on the data shown in Table 8. While the F1 scores show just a minor variance, there is a slight improvement in performance as the time gap is extended. Variables with a longer time lag may be more effective in forecasting financial trouble. We think this is because the six-month time frame is not yet appropriate. The PD, EAD, LGD, and RWA are typically reassessed annually. Hence, a minimum time frame of one year is required to integrate patterns associated with these variables into the model. Testing this association is not feasible due to limited data. For future research, it would be intriguing to examine the correlation between the time gap and model performance using a time span of a minimum of twelve months.

5.6. Watchlist trigger metrics

The financial distress prediction model seeks to establish a novel WL trigger designed to signal when a client may require watchlist

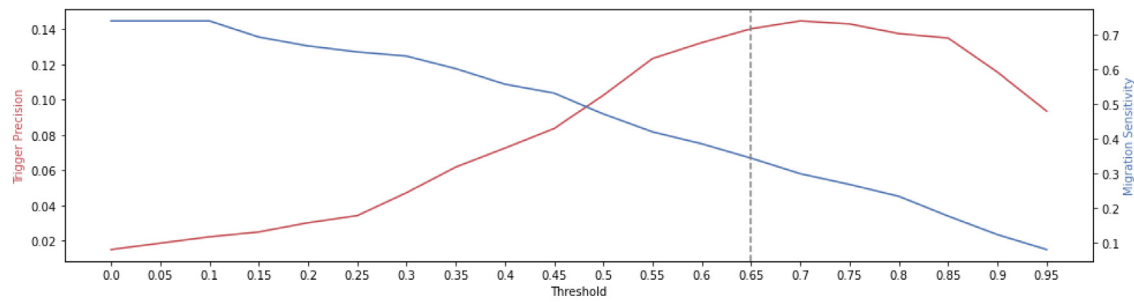


Fig. 4. Trigger precision and migration sensitivity for different threshold values.

consideration. The efficacy of this WL trigger is evaluated through metrics such as trigger precision, migration sensitivity, and time lag. A comparative analysis between June and September 2022 demonstrates that most supervised models, with the exception of XGB and RF, fall short of surpassing the existing triggers in terms of accuracy and sensitivity.

RF, for instance, exhibited a trigger precision of 12.7%, indicating that out of 9474 instances where the trigger was activated, 1202 accurately forecasted a negative migration within the subsequent six months. Its migration sensitivity reached 67.6%, capturing 393 out of 581 negative migrations. In general, comparing the performance of the existing triggers and WL triggers, we acknowledge the superior predictive performance of the WL triggers in most models.

The calculation of trigger precision closely mirrors the precision metric used for ML model evaluation, albeit with minor variations attributable to Random Under Sampling (RUS). The comparison reveals that while precision remains consistent, the migration sensitivity for WL triggers significantly exceeds that of the ML models, underscoring the WL trigger's enhanced ability to detect negative migrations with just a single activation.

Adjusting the decision threshold offers a mechanism to balance migration sensitivity and trigger precision, influencing the model's conservativeness and, consequently, its precision and sensitivity balance, as depicted in Fig. 4. This adaptability allows for customization based on the relative costs of false positives (FP) and false negatives (FN), with FPs generating additional workload and FNs potentially overlooking clients in distress.

However, the limited historical data constrains comprehensive time lag analysis, necessitating further research to ascertain the triggers' timeliness in detecting negative migrations. Future investigations, with extended data, could yield deeper insights into optimizing the balance between detection earliness and accuracy by fine-tuning model parameters such as the time gap and target window.

5.7. Models' explainability

Fig. 5 illustrates SHAP value analyses for two instances – one not predicted to migrate to the watchlist and another predicted for watchlist classification – using waterfall charts. These charts detail the influence of various features on the model's predictions, categorizing their contributions as either positive or negative towards the final outcome. In the first scenario, factors like the total allocated limit and IFRS stage predominantly negatively affected the watchlist classification, leading the model to predict no migration for the client. Conversely, the second scenario highlights features such as average Probability of Default (PD) and total Risk-Weighted Assets (RWA) making positive contributions towards a watchlist prediction.

Beyond individual case studies, aggregating SHAP values across the dataset offers a broader view of feature impact, as depicted in Fig. 6. This analysis reveals the average absolute SHAP value of each feature, shedding light on their overall significance in shaping the model's predictions. Notably, lagged features related to PD, Days Past

Due (DPD), and Loss Given Default (LGD) emerge as highly influential, underscoring their critical role in the predictive process and highlighting the model's sensitivity to these financial indicators. Such insights not only elucidate the model's reliance on specific features but also emphasize the complex interplay of variables in determining watchlist status.

The literature has largely demonstrate that the integration of SHAP values enhances model interpretability by quantifying the impact of each feature on a given prediction [64,65], and it is not different in this study. For managers, this enables case-level diagnostic insights that go beyond a simple risk score. For example, if a client is flagged as high risk due to recent trigger events, low credit utilization, and declining revenue, SHAP values can clearly show which of these factors contributed most to the risk classification. A relationship manager can use this insight to prepare for a client meeting with targeted questions or suggest specific remedial actions (e.g., revising credit limits, requesting financial statements). Therefore, by embedding SHAP plots into internal risk reporting tools, managers can improve transparency, facilitate internal buy-in, and satisfy supervisory requirements for model explainability.

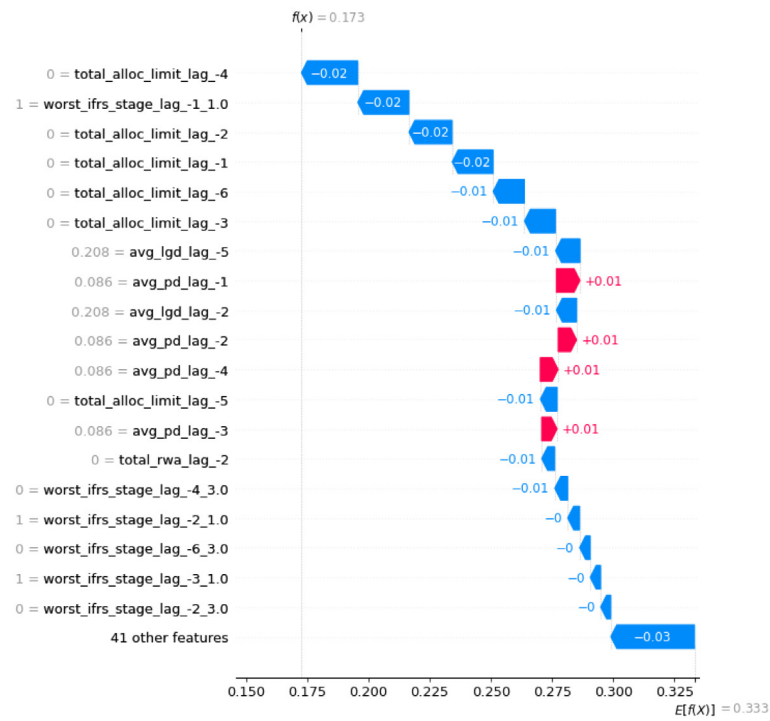
6. Practical considerations

This section reflects on the practical strengths and limitations of the proposed approach, considering its applicability in real-world financial settings. We also discuss key managerial implications, offering actionable insights for risk managers and decision-makers in implementing data-driven early warning systems.

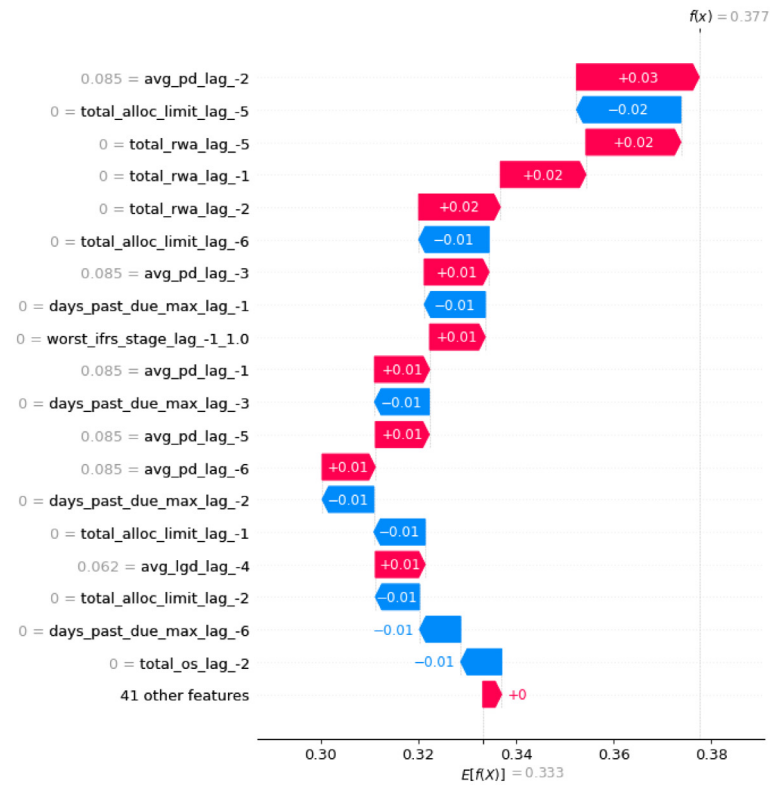
6.1. Strengths and weaknesses of the approach

The proposed approach demonstrates strength in the design and implementation of a robust financial distress prediction model. By integrating internal customer data with early warning trigger information and applying sequential modeling principles, the framework effectively captures the temporal dynamics crucial to identifying clients at risk of negative migration. The use of lagged features, mutual information-based feature selection, and a carefully engineered target window enhances early detection capabilities, aligning well with practical needs in risk management. Among the evaluated models, the RF classifier achieved the highest performance, with strong F1 scores, AUC, and sensitivity—demonstrating its suitability for imbalanced classification tasks. The inclusion of XAI techniques (e.g., SHAP values) further strengthens the model's transparency and interpretability, offering actionable insights for decision-makers. Moreover, the expanding-window cross-validation strategy ensures temporal consistency and realistic evaluation, adding to the methodological rigor of the study.

Despite these strengths, the approach also faces certain practical limitations. The extreme class imbalance in the dataset (only 0.8% representing negative migrations) required the use of Random Under Sampling, which, while computationally efficient, may have led to information loss from the majority class. The performance of neural



(a) Example instance with no negative migration



(b) Example instance with negative migration

Fig. 5. SHAP waterfall plots.

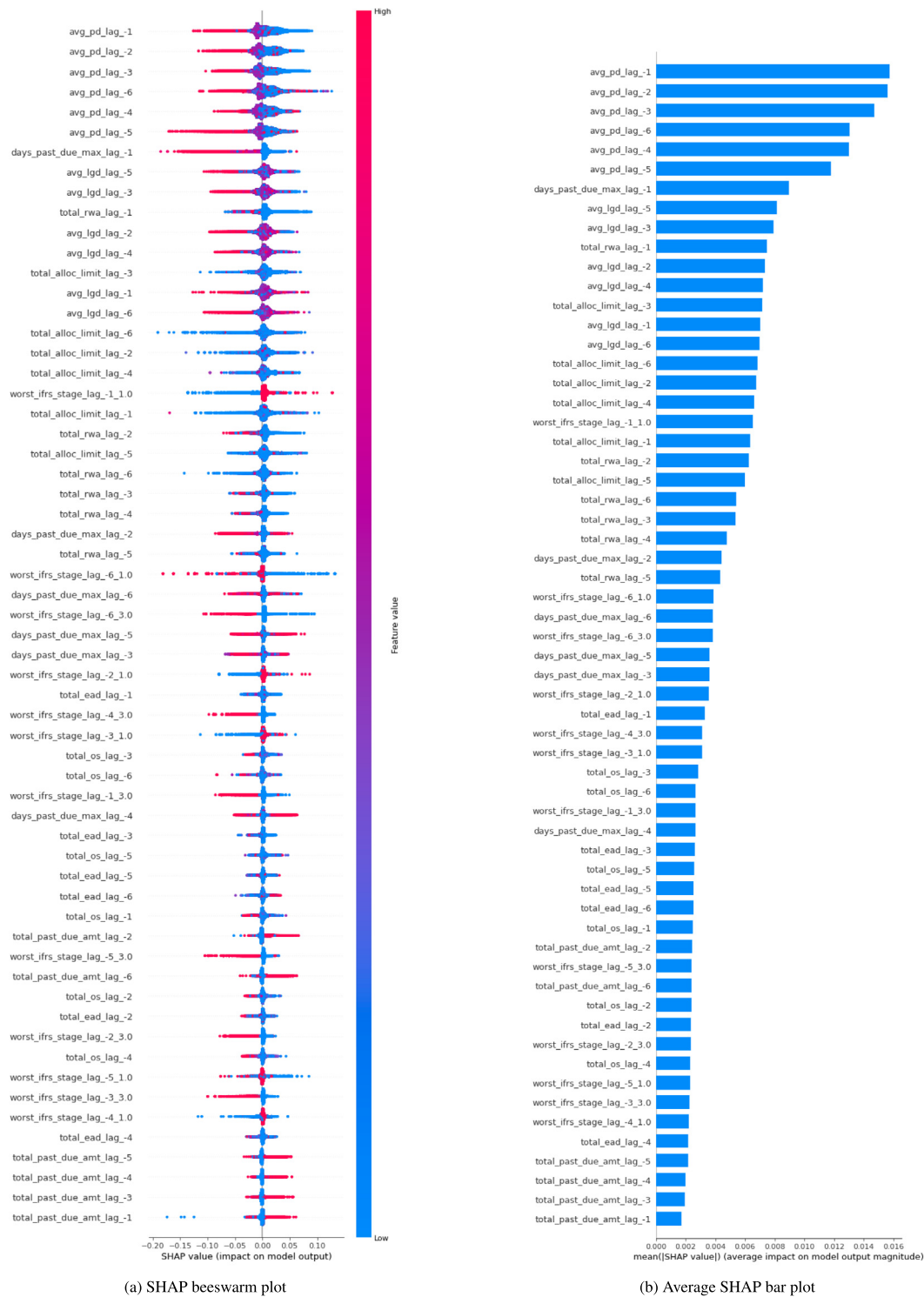


Fig. 6. SHAP value analysis.

networks was notably weaker, potentially due to overfitting on limited and imbalanced data. Additionally, the reliance on a relatively short historical period (January 2021 to September 2022) restricts the generalizability and stability of the models over longer time horizons. These limitations highlight the need for further research using larger, more diverse datasets and alternative resampling or cost-sensitive learning techniques to enhance robustness and operational value.

6.2. Managerial implications

This study offers several implications for industry, especially for risk managers and decision-makers in financial institutions aiming to enhance the early detection of financial distress among commercial clients. By introducing a data-driven EWS that integrates historical trigger events and internal client data, the research highlights the value of predictive analytics in supplementing expert judgment and traditional risk assessment tools. The superior performance of the RF model, particularly in predicting negative migrations with early signals, suggests that financial institutions can adopt such models to more proactively monitor clients and mitigate potential losses.

The findings demonstrate that predictive performance is significantly influenced by strategic modeling choices such as the time window, target window, and decision threshold. Managers should, therefore, consider the trade-offs between early detection and model accuracy, tailoring these parameters based on the specific operational context and risk levels of the institution. The flexibility to adjust thresholds enables institutions to prioritize either precision (minimizing false positives) or sensitivity (capturing more true distress cases), depending on the resource intensity of follow-up actions or the cost of missed risks.

Furthermore, the application of SHAP values contributes to model transparency by revealing the key features driving predictions. This level of interpretability is essential for stakeholder trust and regulatory compliance. Managers can leverage these insights not only to understand the drivers of financial distress but also to design targeted interventions based on the most influential indicators. Overall, the implementation of a model-informed EWS provides a strategic advantage in financial risk management by enabling more timely, explainable, and data-grounded decisions.

Banking managers should formulate a systematic deployment plan to include the suggested EWS into current risk management workflows, thereby translating it into operational impact. This entails synchronizing IT infrastructure to facilitate real-time or periodic model scoring, developing data pipelines for ongoing feature changes (e.g., closely examining customer behavior or macroeconomic indicators), and formulating explicit escalation mechanisms for flagging customers. Furthermore, monitoring protocols must be adhered to in order to guarantee that model performance remains stable over time, despite concept drift or evolving market conditions. This encompasses routine backtesting, periodic recalibration, and governance monitoring via cross-functional risk committees, processes with which financial institutions are already familiar and must adhere to for traditional modeling activities. From a regulatory standpoint, the implementation of ML-based risk instruments must adhere to the openness and fairness criteria established by supervisory bodies. Institutions must preserve audit trails of model decisions, assess predicted fairness among client segments, and engage compliance teams early in the design and implementation phases to reduce governance risks.

7. Conclusion

This research advances the understanding of watchlist (WL) triggers in the context of financial distress prediction within commercial customer settings, offering valuable insights into an increasingly important area for financial institutions. As financial environments grow more complex and volatile, the ability to detect early signs of client distress is critical for mitigating risks and ensuring institutional stability. In this

light, the study addresses a significant gap by providing a structured and data-driven approach to early warning systems (EWS).

Through a case study conducted at a major Dutch commercial bank, the study answers its core research question by analyzing the operational dynamics of early warning triggers and developing predictive models that combine historical event data with internal client information. This dual focus on theoretical rigor and practical relevance underlines the significance of the work in both academic and industry contexts.

Several key contributions emerge from this research. First, the study introduces an innovative analysis of early warning triggers by systematically evaluating their effectiveness across three desirable properties: detectability, probability indication, and timeliness. Using migration sensitivity, trigger precision, and time lag metrics, it offers a nuanced understanding of how these triggers function in practice and influence negative migration predictions. Second, the development of a comprehensive financial distress prediction model represents a methodological advancement. By integrating internal customer data with historical trigger events, and applying a range of supervised learning algorithms, the study showcases the predictive strength of the Random Forest (RF) model in particular. The RF model's F1 score of 0.197 highlights its performance, while additional investigations into time window configurations and the inclusion of SHAP values strengthen the model's robustness and interpretability.

Moreover, the study proposes a strategic implementation of a novel WL trigger, demonstrating its superior performance compared to existing triggers, especially when used in combination with RF and XGBoost models. While limitations in the historical data prevent a full assessment of timeliness, the comparative results affirm the WL trigger's potential for earlier and more accurate identification of financial distress. Another central contribution lies in bridging theoretical and practical gaps by illustrating how internal customer data and early warning triggers can be combined in real-world settings. This application expands the conventional definition of financial distress beyond default to include watchlist classifications, representing a novel contribution to the literature.

In addition to these technical contributions, the research enhances model transparency by applying SHAP values to reveal the key features influencing risk predictions. This not only improves the interpretability of the model but also builds stakeholder trust and supports regulatory compliance. In practical terms, the findings highlight how the WL trigger can improve both the predictive performance and the timeliness of early warning systems in identifying at-risk commercial clients. These insights offer actionable guidance for risk managers in refining and deploying next-generation EWS tools.

The study also introduces new evaluation metrics for early warning triggers and presents a predictive model framework that incorporates a target window concept for early detection—a dimension not yet extensively explored in prior research. Despite its contributions, the study is limited by the historical scope of available data, which constrains the ability to fully assess long-term trigger effectiveness and generalizability. Nevertheless, this limitation presents an opportunity for future research to explore the balance between competing performance metrics, incorporate additional data sources (e.g., macroeconomic indicators or qualitative assessments), and experiment with more advanced machine learning techniques.

The study lays a strong foundation for advancing the early detection of financial distress through data-driven methods. By proposing a roadmap for further exploration, it not only strengthens the current body of predictive modeling but also encourages the adoption of more transparent, adaptable, and sophisticated EWS frameworks in financial risk management practice.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table A.1
Default hyperparameters for each machine learning model.

Model	Default key hyperparameters
DT (Decision Tree)	criterion='gini', max_depth=None, min_samples_split=2
LDA (Linear Discriminant Analysis)	solver='svd', shrinkage=None, priors=None
LG (Logistic Regression)	penalty='l2', C=1.0, solver='lbfgs'
SVM (Support Vector Machine)	C=1.0, kernel='rbf', gamma='scale'
GBM (Gradient Boosting Machine)	n_estimators=100, learning_rate=0.1, max_depth=3
RF (Random Forest)	n_estimators=100, criterion='gini', max_features='sqrt'
XGB (XGBoost)	n_estimators=100, learning_rate=0.3, max_depth=6
ANN2 (Neural Network 2 Layers)	hidden_layer_sizes=(100,), activation='relu', solver='adam'
ANN3 (Neural Network 3 Layers)	hidden_layer_sizes=(100,100), activation='relu', solver='adam'
DUMMY (Baseline Classifier)	strategy='most_frequent'

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Appendix

See Table A.1.

Data availability

The data that has been used is confidential.

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